

CALCITE MATERIALS · CAPABILITY STATEMENT

CORE COMPETENCIES

Supplementary cementitious materials produced by accelerated carbonation of steel slag. Replaces up to 40 percent of Portland clinker with no loss of 28-day compressive strength, cutting embodied carbon by approximately 32 percent per cubic metre of finished concrete.

DIFFERENTIATORS

Our carbonation route consumes CO₂ rather than merely avoiding it, and runs on slag that is currently a disposal liability for steel producers. The process operates at atmospheric pressure, which is what separates it from competing mineralization approaches that require pressurized reactors.

PAST PERFORMANCE

DOE SBIR Phase I (DE-SC0024117) · demonstrated 40 percent replacement at bench scale with ASTM C109 mortar cube strength within 3 percent of the control at 28 days. All Phase I milestones met.

FACILITIES

Pilot line in Dearborn, Michigan producing two tonnes per day, co-located with a materials characterization laboratory including XRD and TGA.